Substitution Procedure

(a) Identify the type of equation and determine the appropriate substitution or transformation.
(b) Rewrite the original equation in terms of new variables. → some ODE which we can solve
(c) Solve the transformed equation.
(d) Express the solution in terms of the original variables.

Ex: $2tx\,dx + (t^2 - x^2)\,dt = 0$

not sep.
not linear
not exact
no sp. int. factor

$$\frac{dx}{dt} = \frac{x^2 - t^2}{2tx}, \quad \boxed{y(t) = \frac{x(t)}{t}} \Rightarrow x(t) = y(t)\cdot t$$

$$\frac{dx}{dt} = \frac{d}{dt}\left(y(t)\cdot t\right) = \frac{dy}{dt}\cdot t + y(t)$$

$$\frac{dy}{dt}\cdot t + y = \frac{y^2 t^2 - t^2}{2t^2 y}$$

$$t\frac{dy}{dt} + y = \frac{(y^2-1)\cancel{t^2}}{2y\cdot \cancel{t^2}} \Rightarrow t\frac{dy}{dt} = \frac{y^2-1}{2y} - y \Rightarrow t\frac{dy}{dt} = \frac{y^2-2y^2-1}{2y} = -\frac{y^2+1}{2y}$$

$$\Rightarrow t\frac{dy}{dt} = -\frac{y^2+1}{2y} \leftarrow \text{separable} \Rightarrow \text{solve find } y(t) \Rightarrow \text{backward substitution } y = \frac{x}{t}$$

Homogeneous ODE

$$\frac{dy}{dx} = f(x,y), \text{ when } f(\alpha x, \alpha y) = f(x,y)$$
$$\alpha \neq 0$$

Substitution: $\boxed{z = \frac{y}{x}}$

$$y = zx, \quad \boxed{\frac{dy}{dx} = \frac{dz}{dx} \cdot x + z}$$

$$\frac{dz}{dx} x + z = f(x, x \cdot z) = f(1, z)$$

↑ depends on z

$$\frac{dz}{dx} = \frac{f(1,z) - z}{x} \leftarrow \boxed{\text{separable}}$$

Ex: $\frac{y^2 - x^2}{2xy} = f(x,y)$

$$f(\alpha x, \alpha y) = \frac{\alpha^2 y^2 - \alpha^2 x^2}{2 \cdot \alpha x \cdot \alpha y} =$$

$$= \frac{\cancel{\alpha^2}(y^2 - x^2)}{2\cancel{\alpha^2} xy} = \frac{y^2 - x^2}{2xy} = f(x,y)$$

Of the form $\frac{dy}{dx} = G(ax+by)$, a, b some real numbers

Subst: $z = ax+by \Rightarrow \frac{dz}{dx} - a = G(z) \leftarrow$ separable

<u>Ex</u>: $\frac{dy}{dx} = y-x-1+(x-y+2)^{-1} = -(x-y)-1+((x-y)+2)^{-1}$

$z = x-y \Rightarrow y = x-z \Rightarrow \frac{dy}{dx} = 1-\frac{dz}{dx}$ $\quad G(s) = -s-1+(s+2)^{-1}$

$1-\frac{dz}{dx} = -z-1+\frac{1}{z+2} \Rightarrow \frac{dz}{dx} = 1+z+1-\frac{1}{z+2} \Rightarrow$

$\Rightarrow \frac{dz}{dx} = z+2-\frac{1}{z+2} \Rightarrow \frac{dz}{dx} = \frac{(z+2)^2-1}{z+2} \leftarrow \boxed{\text{separable}}$

$\frac{1}{2}\int \frac{2(z+2)\,dz}{(z+2)^2-1} = \int dx \Rightarrow \frac{1}{2}\ln|(z+2)^2-1| = x+\ln C$

$u = (z+2)^2$

$du = 2(z+2)\,dz$

$\sqrt{(z+2)^2-1} = Ce^{x}$

$(z+2)^2-1 = C\cdot e^{2x} \Rightarrow (x-y+2)^2 = $

$= C\cdot e^{2x}+1$

Bernoulli equation

$$\frac{dy}{dx} + P(x)y = Q(x)y^n, \quad n \text{ is any real}$$

Subst: $z = y^{1-n} \Rightarrow$ linear DE

Ex: $\frac{dy}{dx} - 5y = -\frac{5}{2}xy^3 \quad (n=3) \qquad z = y^{-2} \Rightarrow y = z^{-\frac{1}{2}}$

$$-\frac{1}{2}z^{-\frac{3}{2}}\frac{dz}{dx} - 5z^{-\frac{1}{2}} = -\frac{5}{2}x\,z^{-\frac{3}{2}} \quad \left(\cdot -z^{3/2}\right) \qquad \frac{dz}{dx} = -2y^{-3}\cdot\frac{dy}{dx}$$

$$\frac{1}{2}\frac{dz}{dx} + 5z = -\frac{5}{2}x \leftarrow \text{linear} \qquad \frac{dy}{dx} = -\frac{1}{2}y^3\frac{dz}{dx} = -\frac{1}{2}z^{-\frac{3}{2}}\frac{dz}{dx}$$

Equations with Linear Coefficients

$$(a_1x+b_1y+c_1)dx+(a_2x+b_2y+c_2)dy=0$$

Subst: $\begin{cases} x=u+h \\ y=v+k \end{cases}$ where (h,k) sol of $\boxed{\begin{cases} a_1x+b_1y+c_1=0 \\ a_2x+b_2y+c_2=0 \end{cases}}$

$\Rightarrow$ homogeneous $\Rightarrow$ separable $\Rightarrow$ solution

Ex: $(-3x+y+6)dx+(x+y+2)dy=0$

$\begin{cases} -3x+y+6=0 \\ x+y+2=0 \end{cases} \Rightarrow -4x+4=0$

$x=1=h$

$y=-3=k$

$\begin{cases} x=u+1 \\ y=v-3 \end{cases}$ $\begin{cases} dx=du \\ dy=dv \end{cases}$

$$\Rightarrow (-3u\cancel{-3}+v\cancel{-3}\cancel{+6})du+(u\cancel{+1}+v\cancel{-3}\cancel{+2})dv=0$$

$$(-3u+v)du+(u+v)dv=0 \Rightarrow \frac{dv}{du}=\frac{v-3u}{u+v}=f(u,v)$$

$$f(\alpha u,\alpha v)=\frac{\alpha v-3\alpha u}{\alpha u+\alpha v}=\frac{v-3u}{u+v}=f(u,v) \leftarrow \underline{\text{homogeneous}}$$